

WORK SHEET - 10

HINTS & SOLUTIONS

Q.1 (B)

Q.2 (B)

Q.3 (A)

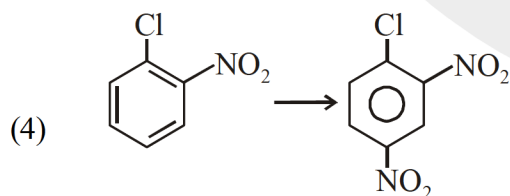
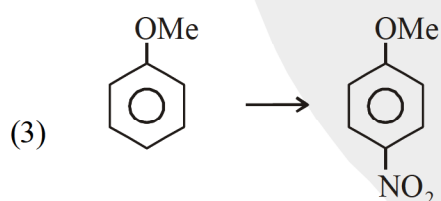
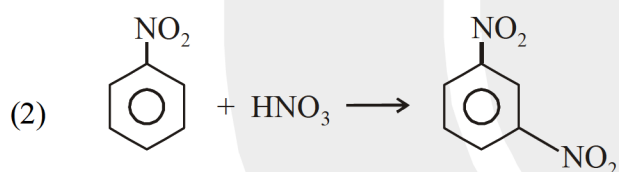
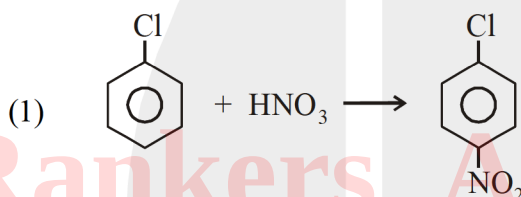
Q.4 (D)

Q.5 (A)

Q.6 (D)

Q.7 (A)

Q.8 (D)



$\downarrow$
-I / -M at o/p position increase rate of $\text{Ar S}_{\text{N}}2$

Q.9 (D)

Q.10 (A) (B) (D)

Q.11 (A) (C)

Q.12 (A) (B) (C)

Q.13 (A) $\rightarrow$ Q; (B) $\rightarrow$ P; (C) $\rightarrow$ S; (D) $\rightarrow$ R

Q.14 (A) $\rightarrow$ P, S; (B) $\rightarrow$ Q, R; (C) $\rightarrow$ Q, S (D) $\rightarrow$ Q, R

Q.15 (A) $\rightarrow$ P; (B) $\rightarrow$ P, Q; (C) $\rightarrow$ R, S; (D) $\rightarrow$ P, Q

Q.16 (A) $\rightarrow$ Q; (B) $\rightarrow$ S; (C) $\rightarrow$ P; (D) $\rightarrow$ R

Q.17 0609

$$pI = \frac{p^{K_{a1}} + p^{K_{a2}}}{2}$$

$$= \frac{9.7958 + 2.39}{2} = 6.09$$

$$p^{K_{a1}} = -\log K_a$$

$$p^{K_{a1}} = -\log 1.6 \times 10^{-10}$$

$$p^{K_{a1}} = 9.79$$

$$p^{K_a} = -\log K_a$$

$$p^{K_{b1}} = -\log K_b$$

$$p^{K_{b1}} = -\log 2.5 \times 10^{-12}$$

$$p^{K_{b1}} = 11.6$$

$$p^{K_{a1}} + p^{K_{b1}} = 14$$

$$p^{K_{a2}} + 11.6 = 14$$

$$p^{K_{a2}} = 2.3979$$

Q.18 0016

Q.19 6524

(a) Acetone $\xrightarrow{2,4\text{DNP}}$ Yellow /orange [6]

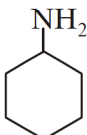
Ethanol $\xrightarrow{2,4\text{DNP}}$ no colour

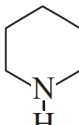
(b) Phenol $\xrightarrow{\text{FeCl}_3}$ Purple colour [5]

1-propanol $\xrightarrow{\text{FeCl}_3}$ no colour

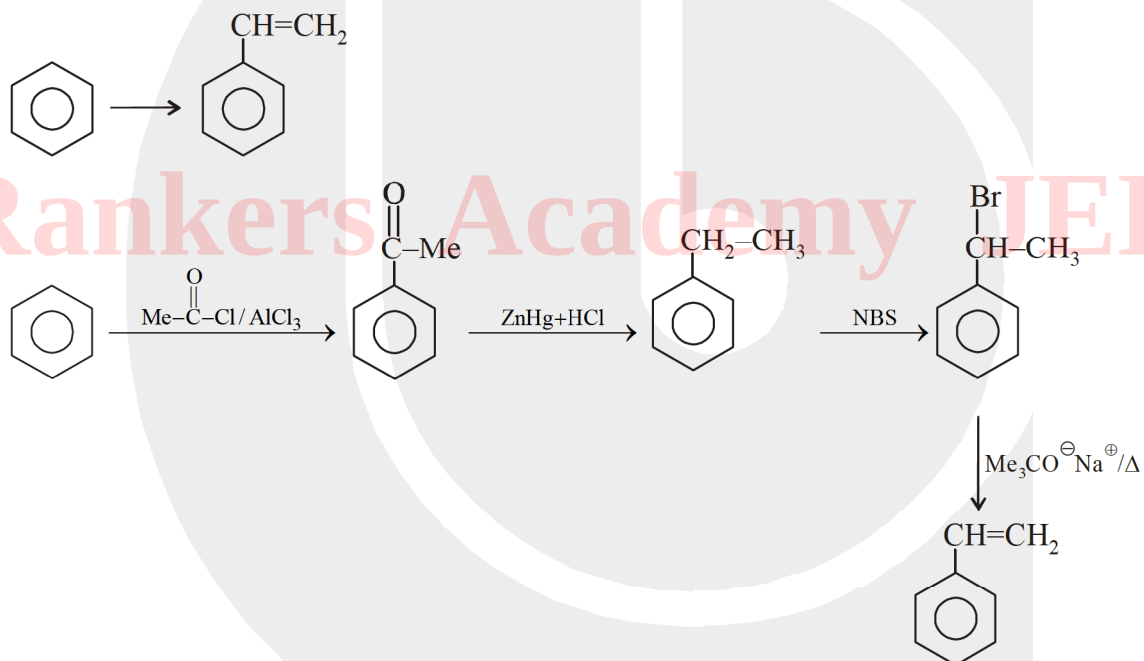
(c) $\text{Me}-\text{CH}_2-\text{C}\equiv\text{CH} \xrightarrow{\text{Tollen reagent}}$ Dirty white ppt. [2]

$\text{Me}-\text{C}\equiv\text{C}-\text{Me} \xrightarrow{\text{Tollen reagent}}$ no ppt.

(d)  $\xrightarrow{\text{Hinsberg reagent}}$ base soluble product [4]

 $\xrightarrow{\text{Hinsberg reagent}}$ base insoluble product

Q.20 7268



Q.21 5573

Number of homopolymer = 5

Number of addition polymer = 7

Number of copolymer = 5

Number of condensation polymer = 3